

# Proximity risk after go-arounds — terminal commands

Every command and option of run\_proximity.py with examples. Run everything from the repository root with the virtual environment active. Method: FRAMEWORK.md.

## 1. Set-up (once per machine)

|                                                                                                                              |                                                                                                                                                                             |
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| <pre>git clone \   https://github.com/sponbob3/CollisionRiskAlgo.git cd CollisionRiskAlgo</pre>                              | Get the repository.                                                                                                                                                         |
| <pre>python3 -m venv .venv source .venv/bin/activate # Windows: .venv\Scripts\activate pip install -r requirements.txt</pre> | Create the virtual environment (Python 3.11 or newer) and install every dependency: the same packages as the go-around pipeline plus scipy, statsmodels, pyproj and pytest. |
| <pre>git pull</pre>                                                                                                          | Update to the latest version later (run inside the repository).                                                                                                             |

## 2. Data

|                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                       |
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| <pre>mkdir -p datasets/KBNA_2025 cp /path/to/daily/*.parquet \   datasets/KBNA_2025/</pre>                         | One file per day (.parquet preferred, .csv accepted) of OpenSky-style state vectors for ALL traffic around the airport, columns: timestamp, icao24, callsign, latitude, longitude, altitude, geoaltitude, vertical_rate, groundspeed, track, onground. The folder name is <ICAO>_<label>; the prefix selects airports/.yaml. datasets/ is gitignored. |
| <pre>python run_proximity.py check-data KBNA_2025 python run_proximity.py check-data \   KBNA_2025 --limit 5</pre> | Data audit only: days found, columns, per-day message and aircraft counts, gaps and receiver outages, altitude references, and whether the volume meets the baseline minimum (28 days, 90 recommended). Writes output/KBNA/proximity_risk/check_data_KBNA_2025.csv.                                                                                   |

## 3. New airport

|                                                     |                                                                                                                                                                                                                                                                                                                                                                               |
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| <pre>python run_proximity.py new-airport KBNA</pre> | Generate airports/KBNA.yaml from the OurAirports database (runways, elevation) with the airspace block pre-filled as field elevation + 4,000 ft. Then EDIT: timezone, preset (air_carrier / training_ga), terrain, assume_arrivals_dataset, airspace.ceiling_ft_msl (verify against the sectional chart), and optionally a flows: block (which runway ends operate together). |
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## 4. Running the analysis

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| <pre>python run_proximity.py KBNA_2025</pre>                                           | The complete run: go-around detection -> output/KBNA/goaround/run_NN/, then baseline (built or reused), encounters, window metrics, every event, sensitivity sweep, tables, figures, per-event plots and summary.pdf -> output/KBNA/proximity_risk/run_NN/ (same NN). |
| <pre>python run_proximity.py KBNA_2025 --limit 5 \   --no-sensitivity --no-plots</pre> | Quick test on the first 5 days without the sweep or figures.                                                                                                                                                                                                          |

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| <code>python run_proximity.py KBNA_2025 --workers 4</code>                                                              | Parallel day processing (default: CPU count - 1).                                                                   |
| <code>python run_proximity.py KBNA_2025 \</code><br><code>--goaround-report --calibrate</code>                          | Also write the go-around stage's figures, narrative report and PDF, and its plateau-duration calibration histogram. |
| <code>python run_proximity.py KBNA_2025 --window 5</code><br><code>python run_proximity.py KBNA_2025 --window 15</code> | Pre/post window length in minutes (default 10; the sweep runs 5 and 15 anyway).                                     |
| <code>python run_proximity.py KBNA_2025 \</code><br><code>--include-ambiguous</code>                                    | Primary event set also includes <code>ga_ambiguous</code> go-arounds.                                               |
| <code>python run_proximity.py KBNA_2025 \</code><br><code>--no-event-plots</code>                                       | Skip the one-PNG-per-go-around plots only (figures and <code>summary.pdf</code> are still produced).                |

## 5. Reusing runs

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| <code>python run_proximity.py KBNA_2025 \</code><br><code>--goaround-run 3</code> | Reuse output/KBNA/goaround/run_03 instead of re-running go-around detection (the proximity run still gets a new NN).                        |
| <code>ls output/KBNA/proximity_risk/cache/2025/</code>                            | Per-day pairwise caches (parameter-hashed). They are reused automatically by every run and sweep; delete the folder to force recomputation. |

## 6. Baselines

|                                                                                                |                                                                                                                                                                                                                                                                                                                                     |
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| <code>python run_proximity.py KBNA_2025 \</code><br><code>--baseline-only</code>               | Build (or reuse) and validate the baseline, write <code>baseline_report.pdf</code> , and stop. Stored in <code>output/KBNA/proximity_risk/baseline/2025/</code> ( <code>baseline.json</code> , <code>baseline_windows.parquet</code> , <code>error_model.csv</code> , <code>phase1_removed.csv</code> , <code>validation/</code> ). |
| <code>python run_proximity.py KBNA_2025 \</code><br><code>--rebuild-baseline</code>            | Force a rebuild even if the stored baseline matches the data and parameters (it is rebuilt automatically when either changes).                                                                                                                                                                                                      |
| <code>python run_proximity.py KBNA_spring26 \</code><br><code>--baseline-from KBNA_2025</code> | Score another dataset's events against the stored KBNA_2025 baseline (same airport).                                                                                                                                                                                                                                                |

## 7. Checks and tests

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| <code>python -m pytest</code>                                                                                                                          | The whole test suite (analytic intervals vs brute force, missed passes, scenarios, pruning, probability vs Monte Carlo, baseline calibration, injected effect, go-around parity, traffic cross-check). About two minutes. |
| <code>python -m pytest -q -m 'not slow'</code>                                                                                                         | Skip the end-to-end synthetic runs.                                                                                                                                                                                       |
| <code>python -m proximity_pipeline.validate_cpa \</code><br><code>KBNA_2025 20250108 20250115</code>                                                   | Cross-check closest-approach distances against the traffic library on the named days.                                                                                                                                     |
| <code>python tools/make_synthetic_dataset.py \</code><br><code>KBNA_synth --days 6</code><br><code>python run_proximity.py KBNA_synth --limit 3</code> | Synthetic traffic for a smoke test without real data ( <code>--intruder-rate</code> and <code>--post-ga-irr</code> inject a known effect).                                                                                |
| <code>python tools/make_terminal_commands_pdf.py</code>                                                                                                | Regenerate this PDF after changing the CLI.                                                                                                                                                                               |

## 8. Where things land

output/KBNA/goaround/run\_NN/

Exact go-around pipeline outputs (all\_approaches.csv, go\_around\_events.csv, summary.pdf, plots/, run\_config.txt).

output/KBNA/proximity\_risk/run\_NN/

run\_config.txt, ceiling\_check.csv, data\_quality.csv, window\_metrics.parquet, encounters.parquet/.csv, go\_around\_risk.csv (+ \_involved), results\_primary.csv, equilibrium.csv, epoch.csv, sensitivity.csv, figures/, events/, summary.pdf.

Options summary: --goaround-run N, --rebuild-baseline, --baseline-only, --baseline-from ICAO\_label, --window MIN, --include-ambiguous, --no-sensitivity, --no-plots, --no-event-plots, --goaround-report, --calibrate, --limit N, --workers N.

Sub-commands: new-airport ICAO, check-data ICAO\_label [--limit N].